

Yixuan Huang

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Research Experience

Postdoctoral Research Associate with Prof. Tom Silver <i>Princeton University</i>	<i>Princeton, NJ</i> Aug 2025 – Present
Visiting Student Researcher with Prof. Jeannette Bohg <i>Stanford University</i>	<i>Stanford, CA</i> Jan 2024 – Aug 2024
Graduate Research Assistant with Prof. Tucker Hermans <i>University of Utah</i>	<i>Salt Lake City, UT</i> May 2021 – Aug 2025
Undergraduate Research Assistant with Prof. Sicun Gao <i>University of California, San Diego</i>	<i>La Jolla, CA</i> Aug 2019 – Aug 2020

Education

University of Utah, Salt Lake City, UT <i>Ph.D. in Computing: Robotics</i> ◦ GPA: 3.98/4.0 ◦ Advisor: Prof. Tucker Hermans	Aug 2020 - Aug 2025
Northeastern University, Shenyang, Liaoning (China) <i>B.E. in Computer Science and Technology (<u>top</u> student in the department)</i> ◦ GPA: 93.2/100, Rank: 1/278	Sep 2016 - Jun 2020
University of California, San Diego, La Jolla, CA <i>Exchange student</i> ◦ GPA: 3.91/4	Sep 2018 - Jun 2019

Publications

*Equal contribution. †Student mentored by me.

Under Review

- [1] **Learning By and For Task and Motion Planning: A Survey**
In Submission, 2026
Yixuan Huang*, Benned Hedegaard*, Naman Shah, Siddharth Srivastava, George Konidaris, and Tom Silver
[Paper]
- [2] **TANDEM: Task and Motion Planning with As-Needed Demonstrations for Efficient Vision-Language-Action Model Fine-tuning**
In Submission, 2026
Samrat Sahoo^{*†}, Liang Ji^{*†}, Tom Silver, and **Yixuan Huang**
[Project Website] [Paper]
- [3] **DATAFARM: Distribution-Aligned Task and Motion Planning for Fine-Tuning Vision-Language-Action Models**
In Submission, 2026
Samrat Sahoo[†], **Yixuan Huang**, and Tom Silver
[Project Website] [Paper]
- [4] **Coding Agents for Generalized Task and Motion Planning Problems**
In Submission, 2026
Matteo Merler, Bowen Li, Josh Roy, Yichao Liang, Qianwei Wang, **Yixuan Huang**, and Tom Silver
[Project Website] [Paper]
- [5] **Chaining Visuomotor Skills for Constrained Manipulation Through Perceptual Manipulation**

In Submission, 2026

Hao Wang[†], **Yixuan Huang**, Somil Bansal, and Tom Silver

[Project Website]

- [6] **SPECTRE: Skeleton Priority Estimation Conditioned on TAMP Refinement Evidence**
In Submission, 2026
Jiacheng Xu[†], **Yixuan Huang**, and Tom Silver
[Project Website]
- [7] **Skeleton-Conditioned Base Placement Evaluation for Long-Horizon Mobile Manipulation**
In Submission, 2026
Qihang Li[†], **Yixuan Huang**, Tom Silver, and Peng Gao

Journal Publications

- [1] **Latent Space Planning for Multi-Object Manipulation with Environment-Aware Relational Classifiers**
IEEE Transactions on Robotics (T-RO), 2024
Yixuan Huang, Nichols Crawford Taylor[†], Adam Conkey, Weiyu Liu, and Tucker Hermans
[Project Website] [Paper]

Conference Publications

- [1] **KinDER: A Physical Reasoning Benchmark for Robot Learning and Planning**
Robotics: Science and Systems (RSS), 2026
Yixuan Huang^{*}, Bowen Li^{*}, Vaibhav Saxena^{*}, Yichao Liang, Utkarsh Aashu Mishra, Liang Ji[†], Lihan Zha, Jimmy Wu, Nishanth Kumar, Sebastian Scherer, Danfei Xu, and Tom Silver
[Project Website] [Paper]
- [2] **RoboVista: Evaluating Vision Language Models for Diverse Robot Applications**
Robotics: Science and Systems (RSS), 2026
Shuangyu Xie^{*}, Kaiyuan Chen^{*}, Ziyang Chen, Simeon Adebola, **Yixuan Huang**, Zehan Ma, Tianshuang Qiu, Wentao Yuan, Dhruv Shah, Pannag R Sanketi, and Ken Goldberg
[Project Website] [Paper]
- [3] **LAP: Language-Action Pre-training Enables Zero-Shot Cross-Embodiment Transfer**
Robotics: Science and Systems (RSS), 2026
Lihan Zha, Asher James Hancock^{*}, Mingtong Zhang^{*}, Tenny Yin, **Yixuan Huang**, Dhruv Shah, Allen Z. Ren, and Anirudha Majumdar
[Project Website] [Paper]
- [4] **Hierarchical DLO Routing with Reinforcement Learning and Multimodal In-Context VLMs**
IEEE International Conference on Robotics and Automation (ICRA), 2026
Best Paper Award on Robot Learning – Finalist
Mingen Li, Houjian Yu, **Yixuan Huang**, Youngjin Hong, and Changhyun Choi
[Project Website] [Paper]
- [5] **Fail2Progress: Learning from Real-World Robot Failures with Stein Variational Inference**
Conference on Robot Learning (CoRL), 2025
Yixuan Huang, Novella Alvin[†], Mohanraj Devendran Shanthi, and Tucker Hermans
[Project Website] [Paper]
- [6] **Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics**
IEEE International Conference on Robotics and Automation (ICRA), 2025
Yixuan Huang, Christopher Agia, Jimmy Wu, Tucker Hermans, and Jeannette Bohg
[Project Website] [Paper]
- [7] **Out of Sight, Still in Mind: Reasoning and Planning about Unobserved Objects with Video Tracking Enabled Memory Models**
IEEE International Conference on Robotics and Automation (ICRA), 2024
Yixuan Huang, Jialin Yuan, Chanh Kim, Pupul Pradhan[†], Bryan Chen, Li Fuxin, and Tucker Hermans
[Project Website] [Paper]
- [8] **Planning for Multi-Object Manipulation with Graph Neural Network Relational Classifiers**

IEEE International Conference on Robotics and Automation (ICRA), 2023
Yixuan Huang, Adam Conkey, and Tucker Hermans
 [Project Website] [Paper]

- [9] **Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots**
International Symposium on Medical Robotics (ISMR), 2021
Best Paper Award – Finalist & Best Student Paper Award – Finalist
Yixuan Huang, Michael Bentley, Tucker Hermans, and Alan Kuntz
 [Paper]

Workshop Papers

- [1] **Learning Multimodal Probabilistic Models of Manipulation Skill Effects**
CoRL 2025 Workshop on Learning to Simulate Robot Worlds
 Siyeon Kim, Mohanraj Devendran Shanthi, **Yixuan Huang**, and Tucker Hermans
- [2] **Fail2Progress: Learning from Failures with Stein Variational Inference for Robot Manipulation**
RSS 2025 Workshop on Out-of-Distribution Generalization in Robotics: Towards Reliable Learning-Based Autonomy
Best Paper Award – Finalist
Yixuan Huang, Novella Alvina[†], Mohanraj Devendran Shanthi, and Tucker Hermans
- [3] **Points2Plans: From Point Clouds to Long-Horizon Plans with Composable Relational Dynamics**
CoRL 2024 Workshop on Learning Effective Abstractions for Planning (LEAP)
Oral Presentation
Yixuan Huang, Christopher Agia, Jimmy Wu, Tucker Hermans, and Jeannette Bohg
- [4] **Latent Space Planning for Unobserved Objects with Environment-Aware Relational Classifiers**
IROS 2023 Workshop on Causality for Robotics
Yixuan Huang, Jialin Yuan, Weiyu Liu, Chanh Kim, Li Fuxin, and Tucker Hermans
- [5] **Reasoning and Planning about Unobserved Objects with Memory Models**
CoRL 2023 Workshop on Neural Representation Learning for Robot Manipulation
Yixuan Huang

Honors and Awards

2026 ICRA Best Paper Award on Robot Learning – Finalist	May 2026
Rising Star at the Princeton Robotics Inaugural Symposium	April 2026
2025 ICRA Doctoral Consortium	May 2025
2021 ISMR Best Paper Award – Finalist	Nov 2021
2021 ISMR Best Student Paper Award – Finalist	Nov 2021
2021 ISMR NSF Travel Award	Oct 2021
University of Utah School of Computing Department Fellowship	Aug 2020
National Scholarship (top 1%)	Nov 2017 & 2018
Northeastern University Excellent Student (top 1%)	Dec 2017 & 2018
Second Place in National Mathematical Modeling Competition in China	Oct 2017
First Place in Provincial Mathematical Modeling Competition	Oct.2017

Invited Talks

Sep 2026, "Towards Robots that Generate Micro-World Models On Demand", IROS 2026 RoBoWoMo Workshop [link], Pittsburgh, PA.

Sep 2026, "Planning Supervises Learning: Like, With, and Beyond Humans for Robot Foundation Models", Host: George Konidaris, Brown Robotics Seminar, Providence, RI.

June 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Rahul Mangharam, University of Pennsylvania, Philadelphia, PA.

Mar 2026, "Toward Structured and Adaptive Robot Physical Reasoning", Host: Felix Heide, Princeton University, Princeton, NJ.

Sep 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", Princeton Postdoctoral Council Seminar Series, Princeton, NJ.

July 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", Oregon State University, Host: Li Fuxin, Corvallis, OR.

May 2025, "Planning for Sequential Multi-Object Robot Manipulation with Learned Relational Dynamics Models", University of Texas at Austin, Host: Roberto Martín-Martín, Austin, TX.

Apr 2025, "From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning", University of Utah Robotics Seminar, Host: Haohan Zhang, Salt Lake City, UT.

Jan 2025, "From Sensor Data to Long-Horizon Plans with Spatial-Temporal Reasoning", University of Utah Graduate Research Symposium, Host: Pratik Soni, Salt Lake City, UT.

Feb 2024, "Reasoning and Planning for Unobserved Objects with Environment-Aware Relational Classifiers and Memory Models", University of Illinois Urbana-Champaign, Host: Katie Driggs-Campbell, Champaign, IL.

Nov 2021, "Toward Learning Context-Dependent Tasks from Demonstration for Tendon-Driven Surgical Robots", University of Utah Robotics Seminar, Host: Jake J Abbott, Salt Lake City, UT.

Teaching

ECE 531/COS 531: Robot Planning Meets Machine Learning

Guest Lecture about Learning Models for Robot Planning

Princeton University

Oct 2025

CS 6300 Artificial Intelligence

Guest Lecture about Reinforcement Learning

University of Utah

Nov 2024

CS 4300 Artificial Intelligence

Guest Lecture about Imitation Learning

University of Utah

April 2022

CS 4300 Artificial Intelligence

Teaching Assistant

University of Utah

Jan 2022 – May 2022

CS 4300 Artificial Intelligence

Teaching Assistant

University of Utah

Aug 2022 – Dec 2022

Mentoring

Mentees who co-authored the publications listed above are marked with [†].

Samrat Sahoo[†] (Princeton Visitor → Stanford Master)

Summer 2026

Hao Wang[†] (Princeton Visitor → Stanford PhD)

Summer 2026

Liang Ji[†] (Princeton Visitor → Stanford Master)

Fall 2025 - Summer 2026

Jiacheng (Joseph) Xu[†] (Princeton Master)

Fall 2025 - Present

Qihang Li[†] (NCSU PhD)

Spring 2026 - Present

Ian Henriques (Princeton Master → MIT PhD)

Fall 2025 - Spring 2026

Tom Wang (Princeton Undergrad)

Fall 2025

Akash Bhowmick (Princeton Undergrad)

Fall 2025

Nichols Crawford Taylor[†] (Utah Undergrad → Northeastern PhD)

Fall 2021 - Spring 2024

Pupul Pradhan[†] (Utah Master → Evans & Sutherland)

Summer 2023

Novella Alvina[†] (Utah Master)

Fall 2024 - Summer 2025

John Chen (Utah Undergrad)

Spring 2025-Summer 2025

Workshop Organization Committees

Learning Effective Abstractions for Planning (LEAP) @ CoRL 2026

Austin, TX

Co-organizers: Naman Shah, Tom Silver, Utkarsh Mishra, David Paulius, Gregory Stein, Yiqing Xu, Amber Li, Franziska Herbert, Georgia Chalvatzaki

Oct 2026

Reviewing

Conference on Robot Learning (CoRL)	2023 - Present
Robotics: Science and Systems (RSS)	2024 - Present
Robotics and Automation Letters (RA-L)	2024 - Present
International Conference on Robotics and Automation (ICRA)	2023 - Present
International Conference on Intelligent Robots and Systems (IROS)	2024 - Present
International Conference on Learning Representations (ICLR)	2025 - Present
Neural Information Processing Systems (NeurIPS)	2026 - Present
IEEE Transactions on Robotics (T-RO)	2025 - Present
IEEE Transactions on Robot Learning (T-RL)	2026 - Present
The International Journal of Robotics Research (IJRR)	2026 - Present
IEEE Transactions on Artificial Intelligence	2024
IEEE Transactions on Instrumentation and Measurement	2024
IEEE Transactions on Industrial Electronics	2024
Workshop on Learning Effective Abstractions for Planning (LEAP @ CoRL)	2024 - Present
RSS Workshop on Out-of-Distribution Generalization in Robotics Towards Reliable Learning-Based Autonomy	2025

Skills

Computer Languages: C/C++, MATLAB, Python (TensorFlow, PyTorch), Java, VHDL

Software & Tools: IsaacGym, ROS, Gazebo, PyBullet, HTML, LaTeX

Outreach

Robotics lab tour co-organizer, University of Utah Bridge Program	2022, 2023
Introduction to mobile manipulator, Undergraduate AI Club at Princeton	2025
Introduction to robot programming for 7th-10th-grade students.	2026